

User Uploads Document (PDF/DOCX)



1. Java Backend
 - Validate File (size, type)
 - Upload to S3/MinIO
 - Publish to Kafka Topic: document-ingestion



2. Kafka (Message Queue)
 - Store Event (file_path, user_id, timestamp)
 - Consumers pick up (Kafka partitions)



3. Python AI Service (Kafka Consumer)
 - Listen to topic
 - Download file from S3
 - Process Document
 - a. Extract text (PDF/DOCX/OCR)
 - b. Clean text (noise removal)
 - c. Chunk (semantic, 500 tokens, 50 overlap)
 - d. Generate Embeddings
 - e. Store in FAISS index
 - f. Save metadata in PostgreSQL
 - Send completion event to Kafka



4. Notification (Optional)
 - Email/Webhook to user: "Document Processed"
 - Update status in database